Hal Hoffmeyer

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EDUCATION

University of Michigan | B.S.E. Computer Science | GPA: 3.97 May 2028
Coursework: Operating Systems, Agentic Software Engineering, Distributed Systems, Machine Learning, Web Systems, Computer Organization, Data Structures and Algorithms, Discrete Math, Statistics and Data Analysis
Awards: James B. Angell Scholar, William J. Branstrom Freshman Prize, University Honors

EXPERIENCE

Cleveland-Cliffs | Software Engineer Intern May 2026 – Present

- Built 9 full-stack web applications (JavaScript, C#, Oracle SQL) that replaced manual, untracked sample handling with a barcode-based logging and tracking system for steel mill lab operations, used daily by 50+ operators to monitor sample types, actions, and process history, significantly reducing costly tracking errors.
- Developed a client-side Web Bluetooth to ZPL bridge with JavaScript that prints QR labels from a browser form directly to on-site Zebra printers, introducing an end-to-end logging system to the team.
- Wrote a C program to detect temperature inversions in real-time sensor data, persisting results to Oracle SQL and automatically triggering alarms for early rust risk detection, reducing 50% of undetected quality defects.
- Engineered a C# server failover system using Oracle SQL row-locking to detect downed servers via job timestamp monitoring, replacing a risky single-server setup with automatic primary/backup redundancy.

Engineering Center for Academic Success | Tutor September 2026 — Present

- Provided one-on-one weekly tutoring to 5+ undergraduate students in 6 computer science courses, including data structures and algorithms, computer organization, and theoretical computer science.

Zeta Pi Tech Fraternity | Full-Stack Web Developer June 2025 — September 2026

- Built a Next.js/Supabase operations portal, managing 50+ events for 100+ users, replacing spreadsheet workflows.
- Created a profile page for students to track their course history, pulling data across 300+ courses via the U-M API and writing a custom React multiselect component to store user records into Supabase.
- Developed a companion course directory by querying user profiles using PostgreSQL and built logic for search, debounce, and filtering features with React hooks.
- Implemented a bug reporting system to auto-file GitHub issues from user submissions using the GitHub REST API.

Detroit Techno Film Web Platform | Web Developer September 2024 — October 2025

- Developed a companion Next.js web app for a documentary, serving 10+ videos and 30+ tracks with NextAuth.js for user authentication, AWS S3 for video storage, and Vercel for database hosting.
- Established the team's Git workflow amongst 8 team members and merged back-end with front-end components.

PROJECTS

AI Pair-Programmer | *Python* September 2026

- Built an LLM-powered pair-programmer CLI tool in Python with an interactive chat, token management, file editing, and Git integration by using http endpoints, SEARCH/REPLACE diff validation, and automated transaction recovery.

Sharded Key/Value Store with Paxos Groups | *Go* April 2026

- Built a fault-tolerant, sharded key/value store in Go with shard migration across Paxos replicated groups.
- Implemented a shard master with atomic handoffs, dynamic shard rebalancing, and duplicate request detection, enforcing linearizability under heavy concurrent workloads via Paxos consensus log serialization.

ICU Patient Mortality Prediction | *Python, NumPy, Scikit-learn, Pandas* February 2026

- Developed a Logistic Regression classifier to predict ICU patient mortality, implementing L1/L2 regularization and extensive hyperparameter sweeps to optimize model selection.
- Engineered a high-dimensional feature pipeline from irregular ICU time-series data, utilizing 5-fold cross-validation and bootstrap sampling to achieve an AUROC of 0.85 on unseen clinical test data.

Languages: C/C++, JavaScript/TypeScript, HTML/CSS, SQL, C#, Go, Python, R, Java
Tools: PyTorch, NumPy, Flask, UNIX, Makefile, AWS, React, Next.js, jQuery, PostgreSQL, Supabase, Git, SVN, Jira, Vim
Certifications: N3 Japanese Language Proficiency Test